

Sampling Stability of Investor-Type Behavioral Coefficients in Aggregate Trading Flows

Symmetric Preference Recovery with Four-Stage Validation

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CCS CONCEPTS

• **Applied computing** → Economics; • **Theory of computation** → Machine learning theory.

KEYWORDS

investor-type trading flows, penalized regression, sampling stability, combinatorial purged cross-validation, behavioral finance

1 METHOD: SYMMETRIC PREFERENCE RECOVERY

SPR is designed to make investor-type comparisons under a common specification. We first define one-day-ahead net-buying behavior, then represent three candidate behavioral channels and two market conditions, and finally estimate the same conditional linear model for every investor type. Four validation procedures test whether the estimated signs persist across resampling schemes, forecast orderings, and estimators. We call this framework *Symmetric Preference Recovery* (SPR). Its coefficients describe conditional associations in aggregate flows, not structural preferences.

1.1 Behavior Forecasting Problem

Let $\mathcal{I} = \{\text{foreign, institutional, individual}\}$ denote the investor types. For type $i \in \mathcal{I}$ on day t , let $V_{i,t}^{\text{buy}}$ and $V_{i,t}^{\text{sell}}$ be the total purchase and sale values. We define normalized net-buying behavior as

$$u_{i,t} = \frac{V_{i,t}^{\text{buy}} - V_{i,t}^{\text{sell}}}{V_{i,t}^{\text{buy}} + V_{i,t}^{\text{sell}}} \in [-1, 1] \quad (1)$$

This ratio scales net purchases by total trading value and preserves both direction and intensity. The prediction target is $a_{i,t+1} = u_{i,t+1}$. The input state $s_{i,t} = (x_{i,t}, C_t)$ contains only information observed through day t .

All investor types share the feature vector $x_{i,t} \in \mathbb{R}^3$, market context $C_t \in \mathbb{R}^2$, and model specification; only their parameters are estimated separately. This symmetry makes the fitted coefficients directly comparable across types without imposing type-specific predictors.

1.2 State Features and Market Context

The state vector $x_{i,t} = (x_t^{\text{mom}}, x_t^{\text{cross}}, x_t^{\text{uw}})^\top$ represents three candidate channels: recent price movement, other types' lagged flows, and an aggregate underwater position.

Recent price movement. Let P_t be the closing price of Samsung Electronics. We use the 20-day log return

$$x_t^{\text{mom}} = \log(P_t/P_{t-20})$$

with a prespecified horizon. A positive (negative) coefficient is consistent with trend-following (contrarian) behavior [1, 3].

Other investor types' lagged flows. For the two types $j, k \neq i$, define $x_{i,t}^{\text{cross}} = (u_{j,t-1} + u_{k,t-1})/2$. Using lagged behavior excludes the mechanical offset among investor flows on the same trading day. This feature captures a lagged cross-type relationship and is not a direct test of within-institution herding [4].

Aggregate underwater position. This feature approximates whether retained inventory is below its recent transaction price. Let $Q_{i,t}^b$ and $Q_{i,t}^s$ denote total purchase and sale quantities, and let $M_{i,t}^b$ and $M_{i,t}^s$ denote their corresponding values. When $Q_{i,t}^b > 0$, the purchase VWAP is $P_{i,t}^b = M_{i,t}^b/Q_{i,t}^b$. In the following recursion, we suppress the common indices (i, t) and write $[v]_+ = \max(v, 0)$:

$$\begin{aligned} \tilde{H} &= \rho H_{t-1}, & H &= [\tilde{H} + Q^b - Q^s]_+, \\ O &= [\tilde{H} - Q^s]_+, & N &= H - O, \\ \bar{P} &= \frac{O\bar{P}_{t-1} + NP^b}{H}, & P^{\text{vwap}} &= \frac{M^b + M^s}{Q^b + Q^s}. \end{aligned} \quad (2)$$

Here, $\tilde{H}$ is decayed inventory entering day t ; O is old inventory remaining after sales; N is newly retained inventory; and $H = O + N$ is total effective inventory. The recursion therefore updates both the retained quantity and its average cost $\bar{P}$, while P^{vwap} summarizes the same-day transaction price. We initialize $H_{i,0} = 0$ and skip any VWAP term whose denominator is zero. We define

$$x_{i,t}^{\text{uw}} = \left[(\bar{P}_{i,t} - P_{i,t}^{\text{vwap}}) / \bar{P}_{i,t} \right]_+.$$

and set it to zero when effective inventory is zero or no trade occurs. The forgetting factor $\rho = 0.98$ gives recent trades greater effective weight and implies a half-life of approximately 34 trading days. This feature is an aggregate proxy rather than an account-level cost basis; its sign can be compared with disposition-effect predictions, but cannot directly identify that effect [5, 6].

Market conditions. Let r_t be the one-day simple return on the KOSPI 200 and e_t the closing KRW/USD exchange rate. Define $z_t^{\text{FX}} = (\log e_t - \mu_{t-1,252}) / \sigma_{t-1,252}$, where $\mu_{t-1,252}$ and $\sigma_{t-1,252}$ are the population mean and standard deviation of $\{\log e_s : s = t - 252, \dots, t - 1\}$. The ordered context vector is $C_t = (r_t, z_t^{\text{FX}})^\top$. Within each split, every feature and context variable is standardized using training indices only and the resulting transformation is applied to the test data. Feature-context correlations range from

−0.29 to 0.22; coefficients are therefore interpreted conditionally, with the remaining terms held fixed.

1.3 Conditional Behavior Model

The model separates average feature sensitivity, context adjustment, and direct context association. For investor type i , the linear score is

$$q_{i,t} = (\beta_i + B_i C_t)^\top x_{i,t} + \alpha_i^\top C_t \quad (3)$$

where $\beta_i \in \mathbb{R}^3$ measures feature sensitivity in the average market context, $B_i \in \mathbb{R}^{3 \times 2}$ captures context-dependent changes in that sensitivity, and $\alpha_i \in \mathbb{R}^2$ captures the direct effect of the context when the features are at their standardized means. Thus, $\beta_i + B_i C_t$ is the effective feature-weight vector at time t . No separate intercept is included.

Let rvec denote row-wise vectorization and define

$$w_{i,t} = [x_{i,t}; \text{rvec}(x_{i,t} C_t^\top); C_t], \quad \theta_i = [\beta_i; \text{rvec}(B_i); \alpha_i]$$

where semicolons denote concatenation. Then $q_{i,t} = \theta_i^\top w_{i,t}$ and $\theta_i \in \mathbb{R}^{11}$. The predicted behavior is $\hat{a}_{i,t+1} = \text{clip}(\hat{\theta}_i^\top w_{i,t}, -1, 1)$. We interpret the five coefficients in β_i and α_i ; the six entries of B_i remain conditioning terms.

1.4 Estimation and Stability Validation

We estimate the common 11-parameter specification using regularization, then test its stability across samples and estimators. Let $\mathcal{D}_i$ denote the training indices for type i . The L_1 -penalized estimator is

$$\hat{\theta}_i = \arg \min_{\theta} \left\{ \frac{1}{|\mathcal{D}_i|} \sum_{t \in \mathcal{D}_i} (w_{i,t}^\top \theta - a_{i,t+1})^2 + \lambda_i \|\theta\|_1 \right\} \quad (4)$$

$\lambda_i \geq 0$ is the type-specific penalty strength. The objective in Eq. (4) is convex, and coordinate descent obtains a global solution. To verify uniqueness, define $n_i = |\mathcal{D}_i|$, $W_i = [w_{i,t}^\top]_{t \in \mathcal{D}_i}$, $y_i = [a_{i,t+1}]_{t \in \mathcal{D}_i}$. For $\lambda_i = 0$, we verify that $\text{rank}(W_i) = 11$. For $\lambda_i > 0$, define the KKT vector $g_i = 2W_i^\top (y_i - W_i \hat{\theta}_i) / n_i$ and set $E_i = \{j : |g_{i,j}| = \lambda_i\}$. We then verify $\text{rank}(W_{i,E_i}) = |E_i|$. The solution is unique for all three investor types in all 45 splits, with a worst-case condition number of 6.86.

We do not interpret a single fitted vector. Instead, each validation procedure targets a different source of instability.

(V1) Combinatorial purged cross-validation. We construct 45 CPCV splits [2] by selecting two of ten chronological folds for testing. We apply a one-day purge and a five-day embargo, and fit every standardization step using training indices only. Because the splits share observations, this procedure measures sensitivity to train–test composition; sign consistency is not interpreted as a p -value.

(V2) Calendar-month block bootstrap. To quantify uncertainty under temporal dependence, we resample calendar-month blocks with replacement 1,000 times and construct 95% confidence intervals. The blocks preserve within-month dependence in daily investor flows.

(V3) Expanding walk-forward evaluation. To enforce strict forecast ordering, we conduct ten expanding-window evaluations because a CPCV test fold may precede some training folds. Starting with 400 observations, each window applies a five-day purge,

Table 1: CPCV out-of-sample performance (mean $\pm$ standard deviation across 45 splits). The metrics summarize predictive association; they do not establish trading value or superiority over alternative forecasting models.

Metric	Foreign	Institutional	Individual
Directional accuracy	0.636 $\pm$ 0.040	0.536 $\pm$ 0.035	0.602 $\pm$ 0.044
Correlation	0.279 $\pm$ 0.106	0.060 $\pm$ 0.062	0.239 $\pm$ 0.116
MAE	0.195	0.095	0.266
Out-of-sample R^2	0.105	0.002	0.102

predicts the next 60 trading days, and expands the training sample by 60 observations.

(V4) Penalty substitution. To test estimator dependence, we re-estimate the model with an L_2 penalty and without a penalty, and then compare coefficient signs across specifications.

We evaluate directional accuracy, prediction–outcome correlation, MAE, and out-of-sample R^2 . Directional accuracy is the proportion of observations for which predicted and realized signs agree, with $\text{sign}(0) = 0$. Out-of-sample R^2 uses the training-sample mean behavior in each split as its baseline prediction. Because behavior dispersion differs across investor types, absolute MAE values are not used to rank them.

2 EXPERIMENTS

2.1 Data

We evaluate SPR in a controlled single-stock setting using the latest 973 daily observations for Samsung Electronics (005930) for which the 252-day exchange-rate history, 20-day momentum, and next-day behavior can all be computed. The state dates span January 6, 2022, to December 29, 2025. Holding the security fixed removes cross-stock differences from the investor-type comparison, although it limits cross-sectional generalizability.

Mean behavior is close to zero for all three types, but its standard deviation is 0.334 for individual investors, 0.257 for foreign investors, and 0.125 for institutions. Their first-order autocorrelations are 0.345, 0.401, and 0.149, respectively. Same-day behavior is strongly negatively correlated across types (foreign-individual: −0.772; institutional-individual: −0.499). The gap between the sum of net purchases by the three types and the market total, which includes other corporations and other foreign entities, averages KRW 14.4 billion per day, or 1.3% of trading value. The type-specific λ values are 0, 0.01, and 0.0003. Because they were carried over from a preceding specification rather than selected by nested cross-validation here, we do not interpret predictive performance as hyperparameter-optimized. Assigning a common $\lambda \in \{0, 0.0005, 0.01\}$ to all types does not alter the sign classifications below.

2.2 Coefficient and Sampling Stability (V1–V4)

The results show a stable foreign–individual contrast but weak institutional predictability. Split dispersion in Table 1 precludes single-split conclusions. Predictions follow broad foreign and individual flow regimes but remain near zero for institutions (Figure 2); Figure 1 shows the corresponding coefficient stability.

Sampling Stability of Coefficients by Investor Type

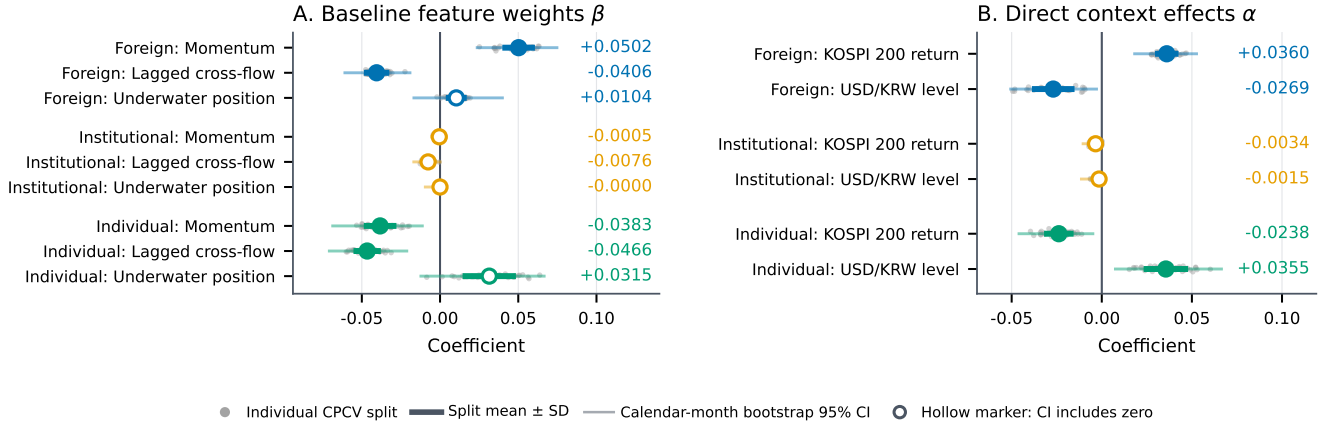


Figure 1: Sampling stability of the recovered coefficients. Gray dots denote the 45 CPCV splits, thick lines show the standard deviation across splits, and thin lines show calendar-month block-bootstrap 95% confidence intervals. Hollow markers indicate intervals containing zero; values on the right are mean coefficients across splits.

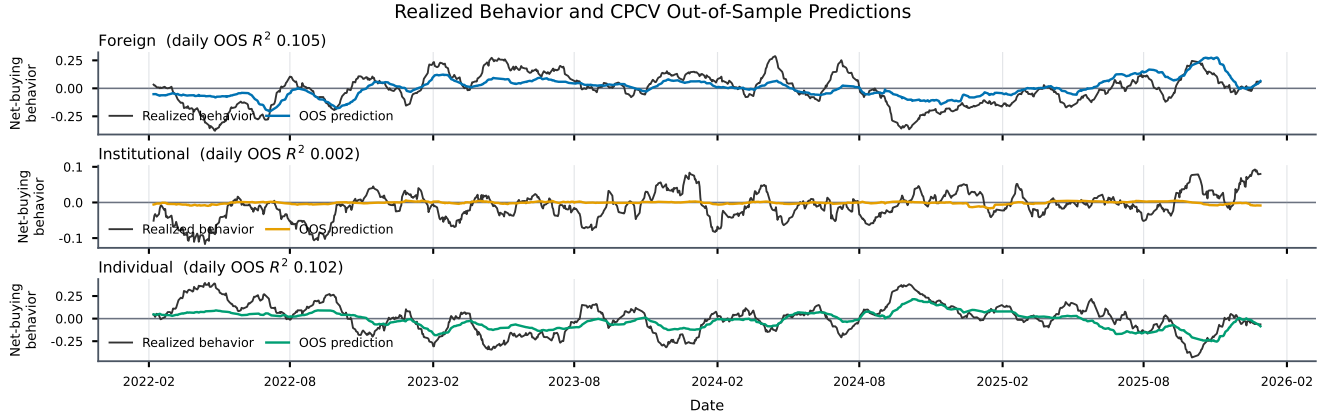


Figure 2: Realized behavior and CPCV out-of-sample predictions by investor type. We first average the nine CPCV test predictions available for each date and then apply a 20-trading-day moving average to both realized and predicted behavior. Panel-level out-of-sample R^2 values are the 45-split means for unsmoothed daily predictions and match Table 1.

(V1) The foreign momentum weight is +0.0502 and positive in all 45 splits, whereas the individual weight is -0.0383 and negative in every split. Lagged cross-flow coefficients are negative for all three types in every split. The individual underwater coefficient is positive in 97.8% of splits. For institutions, the Lasso sets momentum and underwater coefficients exactly to zero in 36 and 44 of the 45 splits, respectively.

(V2) Eight coefficients have bootstrap intervals that exclude zero: momentum, lagged cross-flow, and both direct context effects for foreign and individual investors. The individual underwater interval ranges from -0.0122 to +0.0665, and every institutional interval includes zero. Despite complete sign consistency across the 45 CPCV splits, the individual underwater coefficient

changes sign in 5.5% of calendar-month resamples. Thus, CPCV sign consistency alone understates uncertainty when splits share observations.

(V3) Expanding walk-forward performance varies considerably across windows. Foreign directional accuracy is 0.610 ± 0.116 , and its correlation is 0.131 ± 0.162 , less than half the CPCV correlation. We report this decline as a temporal generalization limit. The coefficient paths are more stable. Foreign momentum remains positive and individual momentum remains negative in all ten windows; the two paths never cross. The same contrast holds for both direct context effects, while the institutional paths remain near zero.

(V4) Foreign and individual coefficients agree to four decimal places under L_1 , L_2 , and unpenalized estimation. Institutional coefficients are more penalty-sensitive; for example, lagged cross-flow changes from -0.0076 to -0.0124 .

Robustness. Across the nine combinations of forgetting factors $\rho \in \{0.95, 0.98, 0.99\}$ and momentum windows $\{10, 20, 60\}$, foreign and individual momentum coefficients are positive and negative, respectively, in 100% of specifications. The individual underwater coefficient is positive in at least 91%. Imposing a common λ across investor types leaves these sign classifications unchanged.

3 INTERPRETATION

- **Foreign investors show stable conditional associations with market direction and the exchange-rate regime.** The momentum weight of $+0.0502$ is positive in all 45 CPCV splits, and its calendar-month bootstrap interval excludes zero. The association between recent price increases and greater foreign net buying is therefore insensitive to the split and resampling scheme. The KOSPI 200 effect of $+0.0360$ and the exchange-rate effect of -0.0269 pass the same checks. Conditional on the other features, foreign net buying is greater on market up-days and lower when the KRW is weak relative to the preceding 252 days. These signs accord with prior evidence on foreign trend trading [1, 3]. Unseparated counterparty flows and common shocks preclude structural or anti-herding interpretations.
- **Individual investors show opposing market associations and a weaker underwater-position signal.** Individual momentum (-0.0383), the KOSPI 200 effect (-0.0238), and the exchange-rate effect ($+0.0355$) have signs opposite to their foreign counterparts. They remain stable across CPCV splits, calendar-month resampling, walk-forward evaluation, and penalty substitution. Conditional individual net buying is therefore lower after stock and market increases and higher during KRW-weakness regimes. Strong negative same-day flow correlation precludes a structural contrarian interpretation. The underwater weight is $+0.0315$ and positive in 97.8% of CPCV splits and at least 91% of robustness specifications. A deeper loss relative to average cost is therefore associated with greater net buying or reduced selling, a sign consistent with avoiding loss realization or averaging down. However, the calendar-month interval includes zero, and aggregate flows cannot distinguish additional purchases from deferred sales. We interpret the coefficient as a limited, repeatedly observed signal rather than direct evidence of loss aversion.
- **Aggregate institutional flows reveal no stable common behavioral weight.** Within the aggregate Korean “institutional” category, the specified features do not form a stable common trading rule. The instability of all main coefficients and an out-of-sample R^2 of 0.002 are consistent with heterogeneous institutional responses offsetting one another during aggregation.

Synthesis. Foreign and individual momentum, market, and exchange rate coefficients point in opposite directions and remain stable across sample constructions. Individual underwater positions retain a limited signal consistent with avoidance of loss realization, whereas aggregate institutional flows reveal no common behavioral rule. The central result is that these differences recur under a symmetric estimation design and four stability checks. Decomposing their origins into independent preferences, counterparty absorption, and common shocks lies outside the identification scope of the analysis.

4 LIMITATIONS AND FUTURE RESEARCH

(1) Interpretive scope. The recovered weights are conditional associations rather than causal effects. The model does not include returns, Sharpe ratios, or transaction costs; Table 1 therefore does not establish the superiority of a trading rule or flow forecast. Because the design does not separately identify the accounting offset in same-day flows and common market shocks, the opposing foreign and individual coefficients and the negative lagged cross-flow coefficients cannot be interpreted as independent preferences.

(2) Disposition-effect proxy. The individual underwater coefficient does not pass the calendar-month block bootstrap. Moreover, the average cost is an approximation based on aggregate trades rather than realized gains and losses at the account level. Its sign is consistent with the disposition effect, but it does not constitute a direct test.

(3) Generalizability. The single-stock setting controls for cross-stock heterogeneity but has not been validated across other securities or markets.

Future research. First, applying the same estimator and V1–V4 procedures to stocks with different foreign ownership and liquidity would test cross-sectional generalizability. Second, distinguishing the accounting offset in simultaneous flows from each investor type’s direct response requires richer data or an alternative identification design. Third, the hypothesis that aggregation produces institutional instability can be tested directly by applying symmetric estimation to the KRX’s disaggregated investor categories.

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